

IILM UNIVERSITY GREATER NOIDA

INDUSTRIAL INTERNSHIP REPORT

Student Performance Prediction & Early Risk Detection Using Machine Learning

Prepared by

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Domain: Data Science and Machine Learning

Project Type: ML Classification & Academic Analytics Prototype

GitHub Repository: [upskillcampus](#)

Code File: StudentPerformancePrediction_Yash_Ramcharan_Chaukse.py

Report File: StudentPerformancePrediction_Yash_Ramcharan_Chaukse_USC_UCT.pdf

This report follows the structure of the USC/UCT sample internship report supplied for the internship.

Executive Summary

This report presents a Data Science and Machine Learning project titled **Student Performance Prediction and Early Risk Detection Using Machine Learning**. The objective is to build a prototype that uses academic and behavioural indicators such as attendance, study hours, assignment performance, internal assessment marks, previous GPA, sleep hours and quiz performance to classify whether a student is likely to remain on track or may require academic support.

The project follows a complete machine-learning workflow: problem definition, data generation, preprocessing, exploratory analysis, feature selection, model training, evaluation and interpretation. Logistic Regression is used as a baseline and Random Forest is used as the main non-linear classifier.

Important note: the dataset used in this demonstration is synthetic and was generated only to make the prototype reproducible. The reported metrics should therefore be treated as a controlled demonstration, not evidence of performance on a real college student population.

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1 Preface

The purpose of this internship project was to gain practical exposure to the way Data Science and Machine Learning can be applied to a real-world decision-support problem. Academic performance is influenced by several measurable factors, and a data-driven system can help identify patterns that may not be obvious from a single examination score.

The selected project, Student Performance Prediction and Early Risk Detection, combines basic data analytics with supervised machine learning. The work was planned as a complete prototype so that every stage—from raw data preparation to model evaluation—could be documented and reproduced.

During the project, emphasis was placed on understanding the data pipeline, selecting appropriate features, comparing a simple baseline with a stronger ensemble model, and interpreting evaluation metrics such as accuracy, precision, recall and F1-score.

Overall experience: the project provided hands-on practice in Python, Pandas, NumPy, scikit-learn, data preprocessing, model training, testing and technical documentation.

Acknowledgement

I express my sincere gratitude to my faculty members, internship coordinators, mentors, and peers who supported me during the project. Their guidance helped me understand the importance of disciplined experimentation, documentation and responsible use of data.

Message to juniors and peers: Start with a small, reproducible problem, understand the data before selecting an algorithm, and always evaluate a model on data that was not used for training.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

According to the supplied USC/UCT sample, UniConverge Technologies Pvt Ltd (UCT) was established in 2013 and works in the Digital Transformation domain, with focus on sustainability and RoI.

The sample describes technologies used by UCT for products and solutions, including Internet of Things (IoT), Cyber Security, Cloud computing (AWS and Azure), Machine Learning, communication technologies such as 4G/5G/LoRaWAN, Java Full Stack, Python and front-end technologies.

The sample also describes UCT IoT Platform (UCT Insight), Smart Factory Platform (Factory Watch), LoRaWAN-based solutions and predictive-maintenance solutions.

2.2 About upskill Campus (USC)

The supplied sample states that upskill Campus, along with The IoT Academy and in association with Uniconverge Technologies, facilitated the execution of the internship process.

The sample describes upskill Campus as a career-development platform focused on personalized executive coaching in an affordable, scalable and measurable way.

2.3 The IoT Academy

The supplied sample describes The IoT Academy as the EdTech division of UCT running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship Program

- Get practical experience of working in the industry.
- Solve real-world problems.
- Improve job prospects.
- Improve understanding of the field and its applications.
- Develop personal skills such as communication and problem solving.

Project Domain: Data Science and Machine Learning

Data Science combines data preparation, statistics, visualization, feature engineering and machine learning to extract useful information from data. For this project, supervised classification is appropriate because each training record has a target label representing the student's outcome category.

2.5 Reference

- scikit-learn documentation and API reference used for model implementation.
- Pandas documentation for tabular data processing.
- NumPy documentation for numerical computation.
- USC/UCT internship report structure supplied in the sample report.

2.6 Glossary

Term	Meaning
DS	Data Science
ML	Machine Learning
EDA	Exploratory Data Analysis
Feature	An input variable used by a model
Target	The output label the model learns to predict
Classification	Predicting a category or class
Precision	Fraction of predicted positive cases that are actually positive
Recall	Fraction of actual positive cases correctly identified
F1-score	Harmonic mean of precision and recall
Random Forest	Ensemble of decision trees used for prediction

Ethical note: Student predictions should never be treated as final judgments. Any real deployment should include consent, privacy controls, bias testing and human review.

3 Problem Statement

Educational institutions collect many indicators of student engagement and achievement, but these indicators are often reviewed separately. A student with falling attendance, low assessment scores and reduced study time may need support before final examination results make the problem obvious.

The problem addressed by this project is therefore: **Can a machine-learning model use structured academic and behavioural indicators to classify whether a student is likely to be on track or may require early academic support?**

Input variables

- Attendance percentage
- Average study hours per day
- Assignment score
- Internal assessment score
- Previous GPA
- Average sleep hours
- Quiz score

Target: Binary outcome: 1 = On Track / Pass and 0 = At Risk. The target in this demonstration is generated from a reproducible synthetic rule with noise and is not derived from real student records.

Expected outcome: The model should estimate the class for a new student record and provide evaluation metrics on held-out data.

4 Existing and Proposed Solution

Existing approach

A common manual approach is to inspect attendance, marks and other academic indicators individually. This can be time-consuming when the number of students increases and may make it difficult to recognize combinations of factors.

Limitations

- Manual review does not scale efficiently.
- Different indicators may be considered inconsistently.
- Early warning may depend heavily on individual judgement.
- Historical patterns are difficult to quantify without a structured model.

Proposed solution

The proposed prototype creates a structured feature table, performs an 80:20 train/test split, trains Logistic Regression and Random Forest classifiers, evaluates them on unseen data, and exposes a simple prediction interface through a Python script.

Value addition

- Repeatable and measurable prediction pipeline.
- Comparison of a baseline model with an ensemble model.
- Multiple performance metrics rather than accuracy alone.
- Clear path for future dashboard or student-support integration.
- Human-in-the-loop design rather than automatic academic decisions.

4.1 Code Submission (GitHub link)

Placeholder: https://github.com//upskillcampus/blob/main/StudentPerformancePrediction_Yash_Ramcharan_Chaukse.py

Replace with your GitHub username after uploading the code.

4.2 Report Submission (GitHub link)

Placeholder: https://github.com//upskillcampus/blob/main/StudentPerformancePrediction_Yash_Ramcharan_Chaukse_USC_UCT.pdf

After uploading the report, replace this placeholder with the actual working report link and update the PDF before final submission.

5 Proposed Design / Model

5.1 High Level Diagram

DATA LAYER	PREPROCESSING	LEARNING LAYER	EVALUATION	DECISION OUTPUT
Attendance Assignment Internal Score GPA Sleep Quiz	Validation Cleaning Feature Preparation	80:20 Split Logistic Regression Random Forest	Accuracy Precision Recall F1-score Confusion Matrix	ON TRACK / AT RISK Human Review

Figure 1: High level flow of the student performance prediction system.

5.2 Low Level Design

Stage	Implementation
Data generation	NumPy distributions + reproducible random seed 42
Validation	Feature ranges and missing-value checks
Target construction	Synthetic weighted score + noise + median threshold
Split	80% training / 20% testing with stratification
Baseline	Logistic Regression
Main model	Random Forest, 250 trees, maximum depth 8
Metrics	Accuracy, Precision, Recall, F1-score, confusion matrix

5.3 Interfaces

- Input interface: CSV/dataframe containing the defined feature columns.
- Processing interface: Python functions and scikit-learn estimators.
- Output interface: predicted class and model evaluation metrics.
- Repository interface: GitHub repository containing source code and report.

A future web interface could accept a validated student record and display the prediction with a confidence score, explanation and recommended human review. Such an interface is outside the current prototype scope.

6 Performance Test

Performance testing is important because a machine-learning model should be evaluated using measurements rather than only visual inspection. For this project, the main constraints are predictive quality, reproducibility and avoidance of misleading conclusions.

6.1 Test Plan / Test Cases

Test Case	Expected Result
Dataset shape check	1,200 rows and 7 input features plus target
Train/test split	80% train and 20% test
Missing values	No missing values after generation
Baseline model	Logistic Regression trains successfully
Main model	Random Forest trains successfully
Metric calculation	All four classification metrics calculated
Repeatability	Fixed seed gives reproducible results
Example inference	A new record receives a binary class

6.2 Test Procedure

- Generate the reproducible demonstration dataset using random seed 42.
- Separate the target column from the input features.
- Perform an 80:20 stratified train/test split.
- Train Logistic Regression and Random Forest.
- Generate predictions on the held-out test set.
- Calculate accuracy, precision, recall and F1-score.
- Inspect the confusion matrix for the Random Forest model.

6.3 Performance Outcome

Metric	Logistic Regression	Random Forest
Accuracy	62.50%	61.25%
Precision	62.30%	61.34%
Recall	63.33%	60.83%
F1-score	62.81%	61.09%

The Random Forest confusion matrix reported for the demonstration is TN=74, FP=46, FN=47, TP=73. The Random Forest model is used as the primary prototype model because it can capture non-linear relationships and interactions between input variables. These values are source-reported demonstration results on synthetic data and should not be interpreted as expected performance on real student data.

7 My Learnings

Technical learning

- Creating and manipulating tabular datasets with Pandas.
- Using NumPy for reproducible numerical data generation.
- Understanding supervised classification.
- Training and comparing Logistic Regression and Random Forest.
- Using train/test splitting and stratification.
- Interpreting accuracy, precision, recall and F1-score.
- Understanding why synthetic results must not be confused with real-world validation.

Professional learning

- Breaking a broad problem into measurable stages.
- Maintaining reproducibility through fixed seeds and documented parameters.
- Writing a technical report that explains design decisions.
- Considering privacy, fairness and human oversight in data-driven systems.

The project also showed that a high metric value alone does not make a model suitable for deployment. Real student data would require careful consent, privacy protection, bias analysis and validation across different groups and academic contexts.

8 Future Work Scope

- Replace synthetic data with anonymized, institution-approved historical data.
- Perform exploratory data analysis and identify missing-data patterns.
- Use cross-validation and hyperparameter tuning.
- Compare additional models such as XGBoost, SVM and calibrated ensembles.
- Add probability/confidence estimates rather than only a class label.
- Build a dashboard for faculty or student-support teams.
- Add model explainability using feature importance and SHAP-style analysis.
- Monitor model drift as student behaviour and academic policies change.
- Evaluate fairness and error rates across relevant groups before deployment.
- Integrate alerts with a human review process instead of automatic action.

Deployment considerations

A production system should minimize the amount of personal data collected, apply role-based access controls, protect stored data, document the model version, and provide a mechanism for students or staff to challenge or review predictions.

9 Appendix: Implementation Summary

Feature	Type	Range / Example
attendance	float	35–100 %
study_hours	float	0.5–9 hours/day
assignment_score	float	20–100
internal_score	float	15–100
previous_gpa	float	3.5–10
sleep_hours	float	4–10 hours/day
quiz_score	float	15–100
passed	integer	0 = At Risk, 1 = On Track

Reproducibility Checklist

- Random seed: 42
- Dataset size: 1,200 synthetic records
- Test size: 20%
- Primary model: Random Forest
- Baseline: Logistic Regression
- Evaluation: Accuracy, Precision, Recall, F1-score and confusion matrix
- Source code: supplied separately as StudentPerformancePrediction_Yash_Ramcharan_Chaukse.py

The accompanying Python file is self-contained and is intended to reproduce the demonstration workflow without downloading a proprietary dataset.

10 Conclusion

The Student Performance Prediction and Early Risk Detection project demonstrates a complete Data Science and Machine Learning workflow for a practical academic-support use case. The project defines a classification problem, prepares structured data, trains baseline and ensemble models, evaluates them on unseen data and documents the limitations of the prototype.

The Random Forest model provides the main prediction component in the demonstration, while Logistic Regression provides a simpler baseline for comparison. The measured results show that the prototype can perform the intended classification task on its controlled synthetic dataset.

Final limitation: because the data is synthetic, these results cannot be used to claim that the model will predict real student performance with the same accuracy. The next major step is validation on anonymized, approved real-world data followed by fairness, privacy and usability testing.

Student Declaration

I, Yash Ramcharan Chaukse, prepared this Data Science and Machine Learning project and report as an academic internship deliverable. The prototype and documentation are intended for educational and demonstration purposes.

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Domain: Data Science and Machine Learning

Project: Student Performance Prediction and Early Risk Detection Using Machine Learning

Signature / Mentor Approval: _____